Application of the Equality Principle to Quality Control in Scientific Knowledge Production

SCX

——M

SCX and Scientific Peer Review as Potential Energy Audit: Reviewer Count M and Coordinate-System Alignment as Gauge Conditions for Scientific Knowledge Production

— SCX —

— An Audit Theory of Peer Review Systems via the SCX Equality Principle —

SCX

July 2, 2026

Abstract

. ——$M = 23$——SCXPotential Energy Audit System$\mathcal{S}_{\text{review}}\mathcal{C}_r$(1) Thm 1$P_{\text{detect}} \to 1M \to \infty M = 2, 3$0.4–0.6——(2) Thm 2$\mathcal{C}_r\mathcal{C}_a\theta_{\text{mis}} > \theta_{\text{crit}}$——Hidden Rejection Tax(3) Thm 3——SCX Thm 3$M \to \infty$Coordinate-System Declaration Protocol$\{\mathbf{e}_i\}\mathbf{g}$——$\sum_m \mathbf{g}_m = \mathbf{0}$

English Abstract. Scientific peer review is the core quality-control mechanism of knowledge production, yet its current structure — relying on merely $M = 2$ or 3 reviewers, typically sharing the same disciplinary training background — suffers from a fundamental audit-theoretic deficiency under the SCX Equality Principle framework. We reformulate peer review as a **Potential Energy Audit System**: each manuscript occupies a position on the review potential surface $\mathcal{S}_{\text{review}}$, and reviewers evaluate its potential value according to their own coordinate systems $\mathcal{C}_r$. We establish three core theorems: (1) **Detection Probability Theorem** (Thm 1): under independent reviewers with diverse coordinate systems, the error-detection probability satisfies $P_{\text{detect}} \to 1$ as $M \to \infty$; yet the current system's $M = 2, 3$ implies detection probability in the 0.4–0.6 range — making the **replication crisis a mathematically expected outcome**. (2) **Coordinate-System Misalignment Theorem** (Thm 2): when the angle θ_{mis} between a reviewer's coordinate system $\mathcal{C}_r$ and the author's coordinate system $\mathcal{C}_a$ exceeds a critical threshold θ_{crit}, the probability that a manuscript is falsely rejected due to coordinate-system mismatch rather than quality failure is positive with a non-trivial lower bound — this constitutes a **Hidden Rejection Tax**. (3) **Single-Reviewer Unidentifiability Theorem** (Thm 3): a single reviewer cannot distinguish between *honest error* and *research misconduct* from observational data alone — a direct corollary of SCX Thm 3 (The Honest Person Theorem) applied to peer review. Solutions follow from the mathematical structure of potential energy audit: $M \to \infty$ via open science (open peer review, post-publication review, independent replication as independent audit), reviewer coordinate-system diversity as an audit-independence condition, and continuous post-publication review as continuous audit. We propose the

Coordinate-System Declaration Protocol: every reviewer must declare, within their review report, the basis vectors $\{\mathbf{e}_i\}$ and gauge posture $\mathbf{g}$ of their disciplinary coordinate system, enabling editors to compute explicit gauge alignment among reviewers — this operationalizes the Equality Principle gauge condition $\sum_m \mathbf{g}_m = \mathbf{0}$ in peer review.

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1

1.1 M

"2323$M = 2$——M"

(Review Note): *This is not a metaphor. This is a formal audit-theoretic claim. The mathematical structure of peer review — M evaluators, each operating from a coordinate system $\mathcal{C}_r$, assessing a target under an unknown ground truth — is structurally identical to a multi-auditor detection problem. The replication crisis, viewed through this lens, is not a sociological accident. It is a* ***Theorem 1 prediction****.*

Replication Crisis————36%–47% [6]11% [7] [8, 9]inter-rater reliability [10]reviewer bias [11]90%

[12]——

publication bias [13]questionable research practices$p$-hacking [14] [15]

Potential Energy Audit System$M$Gauge Alignment——$M = 2, 3$

1.2

M—— [16]SCX

- **(Sufficiency)** $\leftrightarrow$ $MM1$ [3]
- **(Appropriateness)** $\leftrightarrow$ $\sum_m \mathbf{g}_m = \mathbf{0}$——

- ——Cognitive Conflict of Interest
- $M = 2, 3$
- ——Thm 3

1.3 M

$p0 < p < 1$——$M_{\text{eff}}M$

M

$$P_{\text{detect}} = 1 - (1 - p)^{M_{\text{eff}}}. \tag{1}$$

$p \approx 0.4M_{\text{eff}} \approx 2M = 3P_{\text{detect}} \approx 0.64p \approx 0.25P_{\text{detect}} \approx 0.44$——

——$Mp < 0.005p < 0.05$——σM

1.4 SCX

SCX [1, 2]

SCX (SCX Equality Principle)

$$\boxed{\sum_m \mathbf{g}_m = \mathbf{0}}$$

$\sum_m \mathbf{g}_m = \mathbf{0}$gauge cancellation

1.5

23Thm 1Thm 2Thm 34M5$g \neq 0$6$M \to \infty$78

2

2.1

Definition 2.1 (, **Submission Space**). $\mathcal{X}$——$x \in \mathcal{X}\mathcal{X}$

Definition 2.2 (, **Review Potential Surface**).

$$\mathcal{S}_{review} : \mathcal{X} \to \mathbb{R}, \tag{2}$$

$\mathcal{S}_{review}(x)$

(i) scientific quality——

(ii) reviewer assessment——M

(iii) editorial decision——

(iv) post-publication impact——

(Review Note): *Like all potentials in the SCX framework, $\mathcal{S}_{review}$ is a* ***socially constructed*** *potential. It exists because the scientific community agrees (or is institutionally compelled to agree) that certain scientific contributions occupy "higher" positions than others. This is precisely what makes it a* ***gauge-dependent*** *quantity — it is defined only up to the choice of evaluation coordinate system.*

Definition 2.3 (, **Review Potential Gradient**). x

$$\nabla \mathcal{S}_{review}(x) = \left(\frac{\partial \mathcal{S}_{review}}{\partial x_1}, \ldots, \frac{\partial \mathcal{S}_{review}}{\partial x_d} \right)(x), \tag{3}$$

$d = \dim(\mathcal{X})|\nabla \mathcal{S}_{review}(x)|$ ""

(Review Note): *The gradient $\nabla \mathcal{S}_{review}$ is the mathematical expression of what reviewers and editors* ***value****. When a discipline's gradient emphasizes novelty over robustness, the system produces novel but fragile results. When it emphasizes theoretical elegance over empirical grounding, the system produces beautiful but disconnected theory. The gradient is the incentive structure.*

2.2

Definition 2.4 (, **Reviewer Coordinate System**). $r\mathcal{C}_r = (\mathcal{B}_r, \mathbf{o}_r, \Lambda_r, \mathbf{g}_r)$

- $\mathcal{B}_r = \{\mathbf{e}_1^{(r)}, \ldots, \mathbf{e}_{d_r}^{(r)}\}$ *evaluation basis*——
- $\mathbf{o}_r \in \mathbb{R}^{d_r}$ *coordinate origin*—— ""
- $\Lambda_r = \mathrm{diag}(\lambda_1^{(r)}, \ldots, \lambda_{d_r}^{(r)})$ *dimension weight matrix*——

- $\mathbf{g}_r \in \mathbb{R}^{d_r}$ *gauge posture——*

Definition 2.5 (, **Author Coordinate System**)**.** *a*$\mathcal{C}_a = (\mathcal{B}_a, \mathbf{o}_a, \Lambda_a, \mathbf{g}_a)$*——""*

(Review Note): *The mismatch between* $\mathcal{C}_r$ *and* $\mathcal{C}_a$ *is not a flaw to be eliminated — it is an irreducible feature of scientific pluralism. A Bayesian statistician and a frequentist econometrician will evaluate the same paper against incommensurable bases. The question is not whether mismatch exists, but whether the review system* ***accounts*** *for it.*

Definition 2.6 (-, **Reviewer-Author Misalignment Angle**)**.** *ra*

$$\theta_{mis}(r, a) = \arccos\left(\frac{\langle \mathbf{g}_r, \mathbf{g}_a \rangle}{\mathbf{g}_r \cdot \mathbf{g}_a}\right), \tag{4}$$

$\mathbf{g}\theta_{mis} = 0\theta_{mis} = \pi/2$——$\theta_{mis} = \pi$——

2.3

Definition 2.7 (, **Audit Force**)**.** *Mx*

$$\mathbf{F}_{audit}(x) = \frac{1}{M}\sum_{r=1}^{M} \mathbf{r}_r(x), \tag{5}$$

$\mathbf{r}_r(x) \in \mathbb{R}^{d_r} rx$——

Definition 2.8 (, **Review Gauge Condition**)**.** *MReview Gauge Condition*

$$\sum_{r=1}^{M} \mathbf{g}_r = \mathbf{0}, \tag{6}$$

SCX$\sum_m \mathbf{g}_m = \mathbf{0}$

Remark 2.9. (6)$\mathbf{g}_r \equiv \mathbf{g}_0 \sum \mathbf{g}_r = M\mathbf{g}_0 \neq \mathbf{0}\mathbf{g}_0 \neq \mathbf{0}$————

(Review Note): *""——SCX——*

Definition 2.10 (, **Effective Independent Reviewer Count**)**.** $M\Sigma_g \in \mathbb{R}^{M\times M}\Sigma_g^{ij} = \mathrm{Cov}(\mathbf{g}_i, \mathbf{g}_j)$

$$M_{\mathit{eff}} = \frac{\left(\sum_{i=1}^{M} \lambda_i\right)^2}{\sum_{i=1}^{M} \lambda_i^2}, \tag{7}$$

$\{\lambda_i\}\Sigma_g\Sigma_g = \sigma^2 I M_{\mathit{eff}} = M M_{\mathit{eff}} = 1 M_{\mathit{eff}} M$——$M_{\mathit{eff}} \in [1.2, 2.0] M = 3$

3

3.1 1——M

Theorem 3.1 (, **Detection Probability Theorem**)**.** *xfatal flaw——r*$\mathcal{C}_r p_r \in (0,1) M_{\mathit{eff}}$

$$P_{detect}(M_{\mathit{eff}}, \{p_r\}) = 1 - \prod_{r=1}^{M_{\mathit{eff}}}(1 - p_r). \tag{8}$$

(i) **.** $M_{eff} \to \infty \inf_r p_r > 0 P_{detect} \to 1$

(ii) M**.** M_{eff}

$$P_{detect} \geq 1 - \exp\left(-M_{eff} \cdot \bar{p}\right), \tag{9}$$

$\bar{p} = \frac{1}{M_{eff}} \sum_r p_r$

(iii) **.** $M = 3 M_{eff} \in [1.2, 2.0] p_r \in [0.2, 0.4]$

$$P_{detect} \in [0.36, 0.64]. \tag{10}$$

35%–64%——

(iv) **.** $MM = 4M = 5 M_{eff}/MM$

. (i) $\inf_r p_r > 0 \varepsilon > 0 r p_r \geq \varepsilon P_{\text{detect}} = 1 - \prod_{r=1}^{M_{\text{eff}}}(1-p_r) \geq 1-(1-\varepsilon)^{M_{\text{eff}}} \to 1 M_{\text{eff}} \to \infty$SCX Thm 1 [3]1

(ii) $1 - x \leq e^{-x} x \in \mathbb{R} \prod_r (1-p_r) \leq \exp(-\sum_r p_r) = \exp(-M_{\text{eff}}\bar{p}) P_{\text{detect}} \geq 1 - \exp(-M_{\text{eff}}\bar{p})$

(iii) $P_{\text{detect}} \in [1-e^{-1.2 \cdot 0.2}, 1-e^{-2.0 \cdot 0.4}] = [1-e^{-0.24}, 1-e^{-0.8}] \approx [0.21, 0.55] p_r [0.36, 0.64] M_{\text{eff}}$ qed □

1——p-hacking——$M = 2, 3 M = 2, 3$——

Corollary 3.2 (1, **Replication Crisis as Theorem 1 Prediction**). ——$M_{eff} p < 0.005 p < 0.05$ “”$\Lambda_r \sigma$——$M_{eff} M_{eff} \to \infty$

(Review Note): *Corollary 3.2 is the paper's central polemical claim. Changing significance thresholds is a* ***gauge transformation*** *— reparameterizing the same coordinate system. It changes how strictly we measure, but not how many independent auditors we deploy. A stricter ruler does not replace more rulers.*

3.2 2

Theorem 3.3 (, **Coordinate-System Misalignment Theorem**). *$a q(x) \in [0,1] r$ “”*

$$\hat{q}_r(x) = q(x) \cdot \cos\theta_{mis}(r, a) + \varepsilon_r, \tag{11}$$

$\theta_{mis}(r, a)$ 2.6$\varepsilon_r \sim \mathcal{N}(0, \sigma_r^2) \hat{q}_r(x) > \tau_r \tau_r$

(i) **.** $\cos\theta_{mis} < \tau_r / q(x) q(x) = 1 q(x) < 1 \theta_{mis} \mathbb{P}(\mid q(x)) > 0$——*Hidden Rejection Tax*

$$\alpha_{FR}(q, \theta_{mis}, \tau, \sigma) = \Phi\left(\frac{\tau - q \cdot \cos\theta_{mis}}{\sigma}\right), \tag{12}$$

Φ *CDF*$\theta_{mis} \to \pi/2 \alpha_{FR} \to \Phi(\tau/\sigma)$——$q$

(ii) **.**

$$\theta_{mis\,crit} = \arccos\left(\frac{\tau_r}{q(x)}\right), \tag{13}$$

$\theta_{mis} > \theta_{mis\,crit} \mathbb{E}[\hat{q}_r] < \tau_r$——

(iii) **.** $\cos\theta_{mis} \equiv 1 q(x) < \tau_r \theta_{mis} > 0$ “” “”

Proof. (i) rAccept$\hat{q}_r > \tau_r$Reject$\hat{q}_r = q\cos\theta_{\text{mis}} + \varepsilon_r \varepsilon_r \sim \mathcal{N}(0, \sigma_r^2)$

$$\mathbb{P}(\text{Reject} \mid q, \theta_{\text{mis}}) = \mathbb{P}(q\cos\theta_{\text{mis}} + \varepsilon_r \le \tau_r) \tag{14}$$

$$= \Phi\left(\frac{\tau_r - q\cos\theta_{\text{mis}}}{\sigma_r}\right). \tag{15}$$

$\cos\theta_{\text{mis}} = 0\mathbb{P}(\text{Reject}) = \Phi(\tau_r/\sigma_r)$——$qq < 1\theta_{\text{mis}} > 0\mathbb{P}(\text{Reject}) > 0q > \tau_r$

(ii) $\mathbb{E}[\hat{q}_r] = q\cos\theta_{\text{mis}} q\cos\theta_{\text{mis}} < \tau_r \cos\theta_{\text{mis}} < \tau_r/q\theta_{\text{miscrit}} = \arccos(\tau_r/q)$

(iii) $\cos\theta_{\text{mis}} \equiv 1\hat{q}_r \approx q + \varepsilon_r q < \tau_r q \ge \tau_r \cos\theta_{\text{mis}} < 1$——

qed □

2——bias——“”——

Corollary 3.4. *$M = 3\prod_{r=1}^{M} \alpha_{FR}(q, \theta_{misr}, \tau_r, \sigma_r)\tau = 0.6\sigma = 0.2q = 0.7\theta_{mis} = \pi/4 45\Phi((0.6 - 0.7 \cdot 0.707)/0.2) = \Phi(0.105/0.2) = \Phi(0.525) \approx 0.700.70^3 \approx 0.34$——**34%** “”*

3.3 3——

Theorem 3.5 (, **Single-Reviewer Unidentifiability Theorem**)**.** *$rx\mathcal{D}_x$*

(i) $\mathcal{P}_{honest}$——

(ii) $\mathcal{P}_{fraud}$——

$M = 1\mathcal{P}_{honest}\mathcal{P}_{fraud}$-$(x, \mathcal{D}_x)$

$$\mathcal{P}_{honest}(x, \mathcal{D}_x) = \mathcal{P}_{fraud}(x, \mathcal{D}_x), \quad \forall (x, \mathcal{D}_x) \in \mathcal{X} \times \mathcal{D}. \tag{16}$$

- ***A***——
- ***B***——*A*

Proof. SCX Thm 3 [4]SCX Thm 3$M \ge 1$“” “”

- $\mathcal{P}_{\text{honest}}$= “”——
- $\mathcal{P}_{\text{fraud}}$= “”——

“” “” “”$M = 1$

SCX Thm 3$\eta_{\text{err}}\eta_{\text{fraud}}\eta_{\text{err}} = \eta_{\text{fraud}} = \eta$“-”$\mathcal{A}$

$$\max(\text{Error}_{\text{honest}}(\mathcal{A}), \text{Error}_{\text{fraud}}(\mathcal{A})) \ge \frac{\eta\rho}{2}, \tag{17}$$

$\rho\eta > 0\rho > 0$——

qed □

3“”*$M > 1\tau_r\sigma_r M = 1$*——

Corollary 3.6. *————$\mathcal{P}_{honest}\mathcal{P}_{fraud}$*

4 M

4.1 1

——“” ——“//p-hacking” ——“p”

p——

(Core Claim)
M_{eff}
$= \mathcal{X}$
$= \Lambda_r$
$= M_{\text{eff}}$
P_{detect}

4.2 M

“”ground-truth replicability$\rho_{\text{true}} \in (0,1)$——$P_{\text{detect good}}$$1 - P_{\text{detect bad}}$

$$\rho_{\text{published}} = \frac{\rho_{\text{true}} \cdot P_{\text{detect good}}}{\rho_{\text{true}} \cdot P_{\text{detect good}} + (1-\rho_{\text{true}}) \cdot (1-P_{\text{detect bad}})}. \tag{18}$$

$P_{\text{detect bad}}$——Thm 1$P_{\text{detect bad}} \in [0.36, 0.64]$

Table 1: M_{eff}

M_{eff}	$P_{\text{detect bad}}$	$\rho_{\text{true}} = 0.6$	$\rho_{\text{true}} = 0.7$	$\rho_{\text{true}} = 0.8$	$\rho_{\text{true}} = 0.9$
1.0	0.20	0.484	0.583	0.696	0.833
1.5	0.35	0.569	0.661	0.745	0.857
2.0	0.50	0.667	0.737	0.800	0.878
3.0	0.65	0.774	0.824	0.872	0.928
5.0	0.82	0.892	0.921	0.943	0.968
10.0	0.97	0.980	0.986	0.990	0.995

(Review Note): *Table 1 shows that with current $M_{eff} \in [1.5, 2.0]$, the published replicability rate for $\rho_{true} = 0.7$ is about 66%–74%. To achieve >90% replicability, we need $M_{eff} \geq 5$. The empirical replication rates reported in large-scale replication projects (36%–47% in psychology [6], 11% in cancer biology [7]) suggest ρ_{true} may be even lower in some fields, or detection probabilities are even worse than our estimates — possibly due to reviewer correlation driving M_{eff} below our conservative range.*

4.3 p

$p < 0.05$$p < 0.005$gauge transformation$\Lambda_r$$\sigma$——“”SCX$\mathcal{S}_{\text{review}}$——$M_{\text{eff}}$

$p$$\alpha$$\alpha'$$\tau_r$$\tau_r' = \tau_r + \delta(\alpha, \alpha')$——

- M_{eff}1.2–2.0

- $P_{\text{detect}} P_{\text{detect}} = 1 - \prod(1-p_r) p_r M_{\text{eff}}$
- $\theta_{\text{mis}} > 0 p$

$p 5\sigma p < 0.055 \sigma M_{\text{eff}}$——$M_{\text{eff}} \to \infty$

4.4 SCX

Thm 1 3.2

Criterion 4.1 (, **Structural Prediction of Field Replicability).** *FReview Audit Index*

$$RAI(F) = M_{\mathit{eff}}(F) \cdot \bar{p}(F), \tag{19}$$

$M_{\mathit{eff}}(F) \bar{p}(F) F \rho_F RAI(F)$

- *$M_{\mathit{eff}} M_{\mathit{eff}} > 5$*
- *$M_{\mathit{eff}} M = 2$*
- *M_{eff}/M 1*

(Review Note): *Criterion 4.1 is empirically testable. Fields with strong preprint culture (physics, mathematics, computer science) have de facto higher M_{eff} because the "reviewer pool" includes anyone who reads the preprint. Fields that rely exclusively on traditional journal review with small, homogeneous reviewer panels should show systematically lower replicability.*

5 $g \neq 0$

5.1

deviation——"" "" ""SCX
SCXNon-Zero Gauge Posture

Definition 5.1 (, **Reviewer Bias as Non-Zero Gauge Posture).** *r ""$\mathbf{g}_r$*

$$\mathit{Bias}(r) \equiv \mathbf{g}_r > 0. \tag{20}$$

——SCX$\mathbf{g}_r = \mathbf{0}$ "" ""

$$\sum_{r=1}^{M} \mathbf{g}_r = \mathbf{0}. \tag{21}$$

(Review Note): *This reframing is crucial. The traditional "debiasing" approach tries to make each reviewer unbiased individually ($\mathbf{g}_r \to \mathbf{0}$). The SCX approach accepts that all reviewers are biased ($\mathbf{g}_r > 0$) and instead constructs reviewer panels where biases cancel ($\sum \mathbf{g}_r = \mathbf{0}$). The former requires impossible objectivity; the latter requires compositional diversity.*

5.2

(1) **(Confirmation Bias).** $r\mathbf{g}_r\hat{q}_r(x) = q(x) \cdot \cos\theta_{\text{mis}}(r, a) + \varepsilon_r \cos\theta_{\text{mis}}\theta_{\text{mis}} \approx 0 \cos\theta_{\text{mis}} \approx 1 \theta_{\text{mis}} \approx \pi/2 \cos\theta_{\text{mis}} \approx 0$——

(2) **(Prestige Bias).** [17]$\mathbf{o}_r\mathbf{g}_r$institutional component

(3) **(Gender and Racial Bias).** [18]$\mathbf{g}_r$social-identity component

(4) **(Methodological Bias).** $\mathcal{B}_r$methodological dimension$\lambda^{(r)}_{\text{method}}$

$\mathbf{2}MM = 2\theta_{\mathbf{mis}}(r_1, r_2) \approx 0\theta_{\mathbf{mis}}(r_1, a) \gg 0$————**cognitive monopoly**

5.3

double-blind review [19]$\mathbf{g}_r$

$$\mathbf{g}_r^{(\text{blind})} = \mathbf{g}_r - \mathbf{g}_r^{(\text{identity})}, \tag{22}$$

$\mathbf{g}_r^{(\text{identity})}\mathbf{g}_r^{(\text{theory})}\mathbf{g}_r^{(\text{method})}$——

(Review Note): *Double-blind review removes only the visible component of gauge misalignment — the author's identity. The structural components — theoretical commitments, methodological traditions, disciplinary norms — remain fully exposed in the manuscript itself. A reviewer can still tell, from the methods section alone, whether the paper uses "their" approach or a competing one.*

Proposition 5.2. $\mathbf{g}_r = \mathbf{g}_r^{(id)} + \mathbf{g}_r^{(content)}\mathbf{g}_r^{(id)}\mathbf{g}_r^{(content)}\mathbf{g}_r^{(id)}\mathbf{g}_r^{(content)}\sum \mathbf{g}_r = \mathbf{0}$

$$\sum_{r=1}^{M} \mathbf{g}_r^{(content)} = \mathbf{0}, \tag{23}$$

——

6 $M \to \infty$

6.1

————SCXM_{eff}

Definition 6.1 (, **Open Science as Audit Amplification).** M_{eff}

(i) ——*2–3*

(ii) ——M_{eff}

(iii) ——p_r

(iv) ——$\mathbf{g}_r$

(Review Note): *Note that open science does not merely "increase transparency." In the SCX framework, it changes the fundamental parameter of the audit system: M_{eff}. A paper posted on arXiv with open data has M_{eff} in the hundreds or thousands — every reader is a potential auditor. A paper published behind a paywall in a journal with $M = 2$ anonymous reviewers has $M_{\mathit{eff}} \approx 1.5$.*

6.2

one-time audit eventperiodic audit——
post-publication review, PPRcontinuous audit

Definition 6.2 (, **Post-Publication Review as Continuous Audit).** $t = 0$

$$\mathcal{S}_{review}(x, t) = \mathcal{S}_{review}(x, 0) + \int_0^t \mathbf{F}_{audit}(x, \tau)\, d\tau, \tag{24}$$

$\mathbf{F}_{audit}(x, \tau)\tau\mathcal{S}_{review}(x, t)$——

Theorem 6.3 (, **Detection Convergence under Continuous Audit).** $\lambda\lambda r p_r \bar{p} T$

$$P_{detect}(T) = 1 - \exp(-\lambda T \cdot \bar{p}), \tag{25}$$

$\lim_{T\to\infty} P_{detect}(T) = 1$——

Proof. $[0, T]N(T) \sim \text{Poisson}(\lambda T)$

$$P_{\text{detect}}(T) = 1 - \mathbb{E}\left[(1-\bar{p})^{N(T)}\right] \tag{26}$$
$$= 1 - \sum_{n=0}^{\infty} (1-\bar{p})^n \frac{(\lambda T)^n e^{-\lambda T}}{n!} \tag{27}$$
$$= 1 - e^{-\lambda T} \sum_{n=0}^{\infty} \frac{((1-\bar{p})\lambda T)^n}{n!} \tag{28}$$
$$= 1 - e^{-\lambda T} \cdot e^{(1-\bar{p})\lambda T} \tag{29}$$
$$= 1 - e^{-\lambda T \bar{p}}. \tag{30}$$

$\lambda T \bar{p} \to \infty T \to \infty P_{\text{detect}}(T) \to 1$
qed ∎

(Review Note): *Theorem 6.3 is the mathematical justification for post-publication review. Traditional peer review has fixed T (the review period) and fixed M (the reviewer panel). Post-publication review has unbounded T and unbounded M — guaranteeing eventual detection of any flaw. The practical question is: how long is "eventual"? For high λ (popular, important papers), detection happens quickly. For low λ (obscure papers), detection may take years — but it is guaranteed.*

6.3

M

Definition 6.4 (, **Replication as Orthogonal Audit).** $\mathcal{D}_{orig}\mathcal{M}_{orig}\mathcal{D}_{rep} \perp\!\!\!\perp \mathcal{D}_{orig}\mathcal{M}_{rep}\mathbf{r}_{rep}\mathbf{r}_{orig}$

$$\mathbf{r}_{rep} \perp \mathbf{r}_{orig} \qquad \mathcal{D}_{rep} \perp\!\!\!\perp \mathcal{D}_{orig}, \tag{31}$$

——

Thm 3——

7 $\sum_m \mathbf{g}_m = \mathbf{0}$

7.1

$\sum_m \mathbf{g}_m = \mathbf{0}$Coordinate-System Declaration Protocol, CSDP

Protocol 7.1 (, **Coordinate-System Declaration Protocol (CSDP)).** *Coordinate-System Declaration*

CS1 (Evaluation Basis Declaration)

- *(Theoretical Rigor)*
- *(Empirical Strength)*
- *(Novelty)*
- *(Replicability)*
- *(Interdisciplinary Relevance)*
- *(Methodological Innovation)*

$w_i \in [0, 1] \sum_i w_i = 1$

CS2 (Methodological Tradition Declaration)

- *(Experimental)*
- *(Observational)*
- *(Computational)*
- *(Theoretical)*
- *(Qualitative)*
- *(Mixed Methods)*

CS3 (Theoretical Commitment Declaration)

- */*
- */*
- */*

CS4 (Coordinate-System Self-Assessment)

$$\widehat{\theta_{mis}} = //, \tag{32}$$

"[//] []"

Remark 7.2. CSDP——-

7.2

M

(1) **(Gauge Alignment Matrix)**$\theta_{\text{mis}}(i,j)i,j=1,\ldots,MM\times M\theta_{\text{mis}}(i,j)\approx 0i,j$collective blind spot——

(2) **(Gauge Condition Test)**$\sum_{r=1}^{M}\widehat{\mathbf{g}}_r\sum\widehat{\mathbf{g}}_r\gg 0$——

(3) **(Coordinate-System Gap Detection)**

(4) **(Misalignment Rejection Flag)**$\widehat{\theta_{\text{mis}}}$“”

CSDP——3–5——

7.3

CSDPCSDP

- ——CSDPcognitive conflict of interest
- “. . . . . .”——CSDP
- ——CSDP

CSDP

1: CSDP

2:

3: CSDP

4: CSDPEditorial Manager, ScholarOne

8

8.1

individual virtuestructural design“” “” “”——SCXM_{eff}

- “”$p_r M_{\text{eff}} M_{\text{eff}}=1.5 p_r=0.5 1-(1-0.5)^{1.5}\approx 0.65$
- p——$\mathcal{S}_{\text{review}}$
- M_{eff}——

(Review Note): *This is the paper's most uncomfortable implication for the scientific establishment: the replication crisis cannot be solved within the existing peer review structure. Improving reviewer "quality" through training, checklists, or incentives is a p_r improvement — important, but bounded. The unbounded improvement comes from $M_{\text{eff}}\to\infty$, which requires structural changes to how science is evaluated.*

8.2

$\sum_m \mathbf{g}_m = \mathbf{0}$ “”SCXDemocratic Epistemology [20]

(Democratic Epistemology Principle) SCX$\sum \mathbf{g}_r = \mathbf{0}$ —— $=$ $=$

8.3

1 $MM = 2, 3M \geq 5$

2 $M_{\text{eff}} \approx 1M$

3

4

5

6 “MSCX Thm 1P_{detect}”

8.4

(1) $M_{\text{eff}} p_r M_{\text{eff}}$

(2) $\hat{q}_r = q \cdot \cos\theta_{\text{mis}} + \varepsilon_r$

(3) **CSDP**

(4) $M \to \infty M M \to \infty M$

(5) $\lambda\lambda \approx 0$——

M_{eff}——**Thm 1**——

8.5

——M

“”——M

SCX$\sum_m \mathbf{g}_m = \mathbf{0} M_{\text{eff}} \theta_{\text{mis}}$

$M = 1 M = 2, 3 M \to \infty$——

(Final Word)

“”

$M \sum \mathbf{g}_r$

$M = 2$

$M \to \infty$

$$\boxed{\sum_m \mathbf{g}_m = \mathbf{0}}$$

A ASCX

SCX

Theorem A.1 (SCX Thm 1 — , **Multi-Expert Consistency Noise Detection Guarantee**)**.** *$M\theta$ “”$O(e^{-cM})$ 1$c > 0\theta$ [3]*

Theorem A.2 (SCX Thm 3 — , **The Honest Person Theorem**)**.** *$K \geq 2M \geq 1\mathcal{S}\mathcal{P}_{noise}\mathcal{P}_{hard}$ “” “”$n\mathcal{A}\max(Error_{noise}, Error_{hard}) \geq \eta\rho/2 > 0$ [4]*

Theorem A.3 (SCX Thm 7 — , **Cross-Domain Information Preservation**)**.** *Wasserstein [5]——*

B BM_{eff}

M_{eff}

B.1 B.1

$MNij\rho_{ij}$PearsonSpearman$\mathbf{R} \in \mathbb{R}^{M\times M}\{\lambda_k\}$

$$\widehat{M}_{\text{eff}} = \frac{(\sum_{k=1}^{M} \lambda_k)^2}{\sum_{k=1}^{M} \lambda_k^2}. \tag{33}$$

$\lambda_k\widehat{M}_{\text{eff}} = M\lambda_1 = M\lambda_{k>1} = 0\widehat{M}_{\text{eff}} = 1$

B.2 B.2

$\mathbf{f}_r\mathbf{S}$B.1

$$\widehat{M}_{\text{eff}}^{(f)} = \frac{(\sum_{k=1}^{M} \mu_k)^2}{\sum_{k=1}^{M} \mu_k^2}, \tag{34}$$

$\{\mu_k\}\mathbf{S}$

B.3 B.3

$\rho_{\text{published}}$ (18)$P_{\text{detect bad}} M_{\text{eff}} \bar{p}$
——CSDP

C C

(Coordinate-System Declaration)
CS1 —

- : 0.25
- : 0.30
- : 0.20
- : 0.15
- : 0.10

CS2 —
CS3 —
CS4 —

- (Medium)
- (0.30)

(Review Note): *This template requires approximately 3–5 minutes to complete — less than 1% of the total time typically spent on a thorough review. The marginal cost is negligible; the marginal information gain for editors is substantial.*

References

[1] SCX Working Group. *The SCX Equality Principle: Formal Foundations of Gauge-Conditioned Comparison in Multi-Agent Systems.* SCX Technical Report, 2026.

[2] SCX Mixture-of-Experts Working Group. *(Gauge-Fixing Theory for Potential Function Merging and Expert Fusion).* SCX Technical Report, 2026.

[3] SCX Theory Group. *Theorem 1: Multi-Expert Consistency Noise Detection Guarantee (The Unlikely Lone Genius Theorem).* SCX Supplementary Information S1, 2026.

[4] SCX Theory Group. *Theorem 3: The Honest Person Theorem — Fundamental Unidentifiability of Noise from Difficulty.* SCX Supplementary Information S3, 2026.

[5] SCX Theory Group. *Theorem 7: Cross-Domain Information Preservation.* SCX Supplementary Information, 2026.

[6] Open Science Collaboration. *Estimating the reproducibility of psychological science.* Science, 349(6251):aac4716, 2015.

[7] Errington, T.M., *et al. Investigating the replicability of preclinical cancer biology.* eLife, 10:e71601, 2021.

[8] Camerer, C.F., *et al. Evaluating replicability of laboratory experiments in economics.* Science, 351(6280):1433–1436, 2016.

[9] Camerer, C.F., *et al. Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015.* Nature Human Behaviour, 2(9):637–644, 2018.

[10] Bornmann, L. *Scientific peer review.* Annual Review of Information Science and Technology, 45(1):197–245, 2011.

[11] Lee, C.J., *et al. Bias in peer review.* Journal of the American Society for Information Science and Technology, 64(1):2–17, 2013.

[12] Hojat, M., *et al. Journal acceptance rates: A cross-disciplinary analysis of variability.* Learned Publishing, 27(3):183–192, 2014.

[13] Fanelli, D. *Negative results are disappearing from most disciplines and countries.* Scientometrics, 90(3):891–904, 2012.

[14] Simonsohn, U., *et al. p-curve: A key to the file-drawer.* Journal of Experimental Psychology: General, 143(2):534–547, 2014.

[15] Nosek, B.A., *et al. Scientific utopia: II. Restructuring incentives and practices to promote truth over publishability.* Perspectives on Psychological Science, 7(6):615–631, 2012.

[16] Arens, A.A., *et al. Auditing and Assurance Services: An Integrated Approach.* Pearson, 16th edition, 2016.

[17] Blank, R.M. *The effects of double-blind versus single-blind reviewing: Experimental evidence from The American Economic Review.* American Economic Review, 81(5):1041–1067, 1991.

[18] Wennerås, C., & Wold, A. *Nepotism and sexism in peer-review.* Nature, 387(6631):341–343, 1997.

[19] Snodgrass, R. *Single- versus double-blind reviewing: An analysis of the literature.* ACM SIGMOD Record, 35(3):8–21, 2006.

[20] Anderson, E. *The epistemology of democracy.* Episteme, 3(1-2):8–22, 2006.

[21] Cover, T.M., & Thomas, J.A. *Elements of Information Theory.* Wiley, 2nd edition, 1999.

[22] Seligman, M.E.P. *Learned helplessness.* Annual Review of Medicine, 23(1):407–412, 1972.

[23] Dweck, C.S. *Mindset: The New Psychology of Success.* Random House, 2006.

[24] Vygotsky, L.S. *Mind in Society: The Development of Higher Psychological Processes.* Harvard University Press, 1978.

[25] Ladson-Billings, G. *From the achievement gap to the education debt.* Educational Researcher, 35(7):3–12, 2006.

[26] Bjork, R.A. *Memory and metamemory considerations in the training of human beings.* In: Metacognition: Knowing about Knowing, MIT Press, 1994.